

Abdelaziz Shaarawy

PhD Researcher, Extreme Robotics Lab, University of Birmingham

Email: abdelazizshaarawy@gmail.com | axs2185@bham.ac.uk

GitHub: <https://github.com/zizowasfy> | Webpage: <https://zizowasfy.github.io/>

Research Profile

PhD researcher at the Extreme Robotics Lab (ERL), University of Birmingham, specializing in learning-based motion planning, task generalization, and human-in-the-loop control for robotic manipulation. My research bridges probabilistic learning frameworks and data-driven approaches with state-of-the-art planners for adaptive motion generation in unstructured environments. I have conducted extensive real-world experimentation and validation on single and dual-arm robotic platforms within the Horizon Europe project REBELION, contributing to disassembly of EV batteries applications.

Education

- **PhD in Robotics and AI**, University of Birmingham, UK — 2022–Present
Thesis: *Learning-Based Motion Planning for Human Skill Adaptation in Robotic Manipulation*
 - **MSc in Computer Science (Robotics & Computer Vision)**, Innopolis University, Russia — 2020–2022
Thesis: Haptic Teleoperation and Shared Autonomy for Human–Robot Interaction.
 - **BSc in Mechanical Engineering (Mechatronics)**, Nile University, Egypt — 2015–2020
Graduated with High Honors.
-

Research and Academic Experience

- **Postgraduate Researcher**, Extreme Robotics Lab, University of Birmingham — 2022–Present
Developing learning-from-demonstration motion planners integrating TP-GMM and RRT frameworks. Conducting real-world experimental validation on robotic systems for electric vehicle (EV) disassembly tasks under the REBELION project.
- **Postgraduate Teaching Assistant**, College of Engineering and Physical Sciences, University of Birmingham — 2025–Present
Assisting in undergraduate and master's modules in modelling, design, and programming; potentially supervising student research projects.

- **Lab Researcher**, Laboratory of Intelligent Robotic Systems, Innopolis University — 2021–2022
Worked on several industrial research projects including cable-driven robot teleoperation using several control schemes. Developed control algorithms for precise 3D construction.
 - **Teaching Assistant and Team Lead**, Mechatronics Program, Department of Mechanical Engineering, Nile University — 2018–2019
Assisted in delivering undergraduate modules, supporting both theoretical and laboratory sessions. Led extracurricular student projects and competitive teams focused on applied mechatronics systems development.
-

Research Contributions and Collaborations

- Developed a Task-Parameterized, Sampling-Based Motion Planner adaptable to complex and unstructured environments. The planner integrates probabilistic task representations into sampling-based search for improved generalization and efficiency, and is implemented as part of the **open-source** MoveIt framework. This work was conducted under the **REBELION project**, funded and supported by **UKRI Horizon Europe**.
 - Implemented the developed learning-based planning framework within a Task and Motion Planning (TAMP) system for dual-arm robotic manipulation in an EV battery disassembly real-world scenario. This research was carried out within the **REBELION project** and partially supported by **the Ministry of National Education, Republic of Turkey**.
 - Contributed to a shared autonomy framework integrating Gaussian Mixture Regression (GMR) for adaptive teleoperation and skill transfer. The research was supported in part by the **UKRI, REBELION EU** project, and by the **Royal Society** International Exchanges 2022 Cost Share Grant under the project *“Mutual cross-contamination between automated non-destructive testing and adaptive robotics for extreme environments”*.
 - Designed and implemented two teleoperation control schemes for human–robot interaction and data acquisition. A comparative case study was conducted to evaluate both schemes in performing manipulation tasks within hazardous environments. This work was part of the **REBELION** and **RELIB** projects, both partially funded and supported by **UKRI**.
-

Selected Publications

1. **A. Shaarawy**, A. Rastegarpanah, R. Stolkin, *“Task-aware motion planning in constrained environments using GMM-informed RRT planners,”* **Robotics and Computer-Integrated Manufacturing**, 2026.

2. A. Rastegarpanah, C. Mineo, C. Contreras, **A. Shaarawy**, G. Paragliola, R. Stolkin, “*Haptic Teleoperation in Extended Reality for EV Battery Disassembly Using Gaussian Mixture Regression*,” **Journal of Field Robotics**, 2024.
 3. J. Hathaway, **A. Shaarawy**, et al., “*Toward Reuse and Recycling of Lithium-Ion Batteries: Tele-robotics for Disassembly of Electric Vehicle Batteries*,” **Frontiers in Robotics and AI**, 2023.
 4. K. Almaghout, R. Bobby, M. Othman, **A. Shaarawy**, A. Klimchik, “*Robotic Pick and Assembly using Deep Learning and Hybrid Vision/Force Control*”, **IEEE International Conference ‘Nonlinearity, Information and Robotics (NIR)’**, 2021
-

Awards and Recognition

- 1st Place — *Egypt IoT & AI Challenge (2020)*, Best Robotics Research Project.
 - 5th Place — *Dell Technologies Graduation Project Competition (2020)*.
 - Invited Presenter — *IEEE Global Conference on IoT & AI (GCAIoT 2020)*.
 - 2nd Place — Awarded at Nile University’s Annual Research Forum.
 - Recipient — *College of Metallurgy and Materials PhD Scholarship*, University of Birmingham.
-

Technical Expertise

- **Programming:** ROS (6 years), MoveIt, Python, C++, MATLAB, TensorFlow.
 - **Robotics Platforms:** UR10e, KUKA LBR iiwa, Dual-arm systems.
 - **Simulation & Design:** Gazebo, RViz, Unity3D, SolidWorks.
 - **Research Tools:** Git, LaTeX, OpenAI Gym, Linux/Ubuntu.
-

Future Research and Vision

My long-term vision is to advance the next generation of intelligent robotic systems that safely and adaptively operate in complex, real-world environments. I aim to develop generalizable frameworks that integrate human expertise with AI and motion intelligence to enable resilient robotic decision-making and execution. I aim to continue pursuing my research career as a postdoctoral researcher in the UK, developing sustainable intelligent robotic systems for hazardous industrial applications, fostering translational innovation that ensures robotics research has tangible societal, economic, and environmental impact. I am particularly committed to establishing and expanding national initiatives in AI safety, industrial automation, and field robotics, collaborating with national and international research hubs. Through these efforts, I aspire to strengthen the UK’s global leadership in the domain, driving transformative change toward safer, and sustainable intelligent systems.
